

EFFECTIVE TEACHING OF FEEDBACK ANALYSIS

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ABSTRACT

Feedback analysis is a valuable tool for analyzing multistage feedback amplifiers. Students, however, often find this topic confusing and difficult to learn. Reasons for this difficulty and confusion include applying feedback analysis to inappropriate circuits and failure to explicitly state and check the basic assumptions of feedback analysis. Another source of confusion is failure to keep the source and the load separate from the basic amplifier making the feedback model less physical. In this paper, these pitfalls are discussed and a straightforward methodology for analyzing feedback amplifiers is presented. Several examples of feedback analysis are presented.

I. INTRODUCTION

Before computers, feedback analysis was virtually the only way to analyze multi-stage feedback amplifiers. Today, students and engineers have significant computer power. However, simulator programs such as SPICE typically only give numerical answers that are not very useful for design purposes. Some computer programs are capable of producing symbolic answers. However, for multi-stage amplifiers such solutions are very complicated and not suitable for interpretation. Feedback analysis techniques are still very useful because they yield simple approximate symbolic expressions that are very useful for design. Additionally, proper application of feedback techniques yields considerable insight into the operation of the feedback amplifier.

Consider the general feedback block diagram shown in Figure 1. For feedback analysis, the following assumptions are made. [1]

- 1) The basic amplifier or A network is not loaded by the feedback network.
- 2) The feedback or β network is not loaded by the basic amplifier.
- 3) Both the basic amplifier and the feedback network are unilateral.

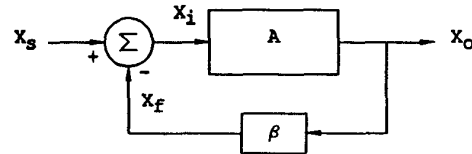


Figure 1. General Feedback Block Diagram.

$$\begin{aligned}X_f &= \beta X_o \\X_i &= X_s - X_f \\X_o &= A X_i\end{aligned}$$

$$A_f = \frac{X_o}{X_s} = \frac{A}{1 + A\beta}$$

Under these assumptions the gain of the feedback system can be found as follows,

where A_f is the gain with feedback, A is the open loop gain and β is the feedback factor. The factor $1 + A\beta$ is often referred to as the desensitivity factor, D . [2]

In electronic amplifiers, the input and output signals are either voltages or currents and, therefore, there are four basic types of amplifiers: voltage, current, transresistance, and transconductance. For feedback amplifiers, either the output voltage or current can be sampled and the feedback signal can be either a voltage or current. Therefore, there are four feedback topologies to consider. Shunt sampling monitors the output voltage at the output node and series sampling monitors the output current in the output loop. Series mixing subtracts the feedback voltage from the input voltage in the input loop and shunt mixing subtracts the feedback current from the input current at the input node.

If the basic assumptions can be approximately met, feedback analysis can be used to analyze the amplifier. The first step is to relate the amplifier to the block

diagram of Figure 1 in order to identify the A and β circuits. Some texts try to apply such analysis to simple circuits such as a one stage amplifier. This, however, creates problems. For example, consider a common emitter amplifier with an emitter resistor. The emitter resistor is common to both the A and β networks making it difficult to separate the basic amplifier and feedback network, and thus, it is difficult to relate this circuit to the feedback block diagram. Furthermore, the basic feedback assumptions are not well met. In one stage amplifiers, the assumptions that both the A and β networks do not load each other and are unilateral are also difficult to satisfy. The final and most compelling reason not to apply feedback analysis to a one stage amplifier is that it can be solved very easily using standard nodal or loop analysis. Using feedback analysis for such a circuit often takes more work to yield an approximate and often not very accurate answer. Students find this very confusing.

Feedback analysis should, therefore, be reserved for multi-stage feedback amplifiers for which it is easier to separate the A and β networks and to satisfy the basic feedback assumptions. Feedback analysis uses a subcircuit to find the open loop gain, A, and a subcircuit to find the feedback factor, β . In a multi-stage amplifier, the open loop gain, A is usually large. If $A\beta \gg 1$, the closed loop gain is approximately $1/\beta$. The closed loop input and output resistances can be found by multiplying or dividing the input and output resistances of the A circuit by the factor D. For voltage (series) mixing, the input resistance is increased and for current (shunt) mixing the input resistance is decreased. For voltage (shunt) sampling, the output resistance is decreased, and for current (series) sampling, the output resistance is increased. It is usually possible to find very simple expressions for the feedback amplifier gain, and input and output resistances. Using these simple approximate analytical expressions from feedback analysis, the student or engineer can determine the components that control circuit performance. The simple expressions can be used to find approximate numerical values, and more accurate values can be found using a simulator such as SPICE. Therefore, it is not important or worth the effort to try to get too accurate of an answer using feedback analysis. The main goal is to get a good approximate relation that gives the functional relations between component values and circuit performance. The approximate relations are very useful for refining a circuit design using a "design through repeated analysis procedure," where values are tweaked during successive simulation runs until desired circuit performance is achieved.

When performing feedback analysis, it is important

to relate the circuit to the feedback block diagram. The feedback topology is identified and the A and β circuits are drawn. It is recommended to keep the source separate from the basic amplifier separate from the load as illustrated in Figure 2. Many texts lump either or both the source load resistance into the basic amplifier. This type of procedure is not only less physical but is quite confusing to the student. Careful and logical parsing of the feedback amplifier into subcircuits to be used to find the open loop gain, A, and the feedback factor, β , is a key step in feedback analysis.

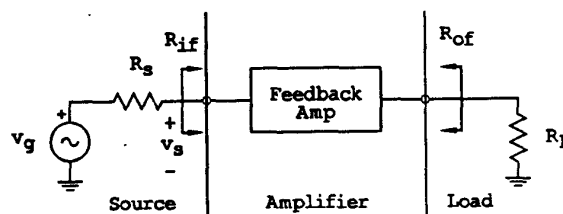


Figure 2. Feedback Amplifier.

To analyze a feedback amplifier, first determine if the output voltage or current is sampled. First short the output voltage and determine if the feedback is removed. If it is, then the output voltage is sampled. For voltage sampling, the input to the feedback network is in parallel with the output voltage. The point where the output voltage is sampled is the output node. If the output voltage is shorted and the feedback is not removed then it is most likely that the output current is sampled. For current sampling, there is typically a resistor in the output loop which is not the load resistor. The function of this resistor is to develop a voltage proportional to the output current. The loop containing the sampling resistor and the load resistor is the output loop. The output voltage sampling node or the output current sampling loop are important as they are the output point in the feedback diagram.

Next, determine whether a feedback voltage or current is mixed. If a feedback voltage is mixed, then the input loop must be identified. Voltages in series add; therefore, the input loop contains the source voltage, the feedback voltage, and the input voltage to the basic amplifier. If a feedback current is mixed, then the input node must be identified. Currents in parallel add so the input node is connected to the source current, the feedback current, and the input current to the basic amplifier. The input voltage mixing loop, or the input current sampling node, are important as are the summation symbol in the feedback block diagram.

After identifying the sampling and mixing, the A and β circuits are drawn. The A circuit is used to determine the gain of the basic amplifier including the loading effects of the feedback network. This gain is referred to as the open loop gain. The β network is used to determine the feedback factor. Loading effects are included to better satisfy the basic feedback assumptions.

The application of feedback analysis is best illustrated by example.

II. EXAMPLE 1

Consider the circuit of Figure 3, which is a noninverting op amp configuration. If the op amp is ideal, feedback analysis is essentially exact since the basic feedback assumptions are nearly exactly met.

It should be noted that the analysis presented below could also be used for a multistage discrete amplifier. Shorting the output voltage removes the feedback, therefore, the output voltage is sampled. The feedback network is comprised of resistors R_1 and R_2 which form a voltage divider that samples the output voltage and feeds back a voltage, V_f , in series with the input signal. Thus, this feedback amplifier is an example of voltage (shunt) sampling and voltage (series) mixing. The output node and input loop are shown in Figure 3. Figure 4 shows the circuits needed to find the open loop gain, A, and the feedback factor, β . To account for the loading effects, a procedure derived from two-port analysis is used. [2]. To account for the loading of the feedback network on the basic amplifier, either short or open the opposite port of the feedback network according to the following rules: (1) If it is series, sever it and (2) If it is shunt, short it. Thus, to account for β network loading on the output of the amplifier, the mixing side is severed at S, shown in Figure 3. To account for the loading of the β network on the input of the amplifier, the sampling side is shorted at the output node.

The gain that should be found is the gain that is optimized for the feedback topology. For our example, the output voltage is sampled and there is voltage mixing at the input. This feedback topology optimizes a voltage amplifier and, therefore, the open loop voltage gain, A_v , needs to be found. The feedback factor for this topology is the ratio of the feedback voltage to the output voltage.

Next, solve the circuits of Figure 4 to find A, β , the input and output resistances.

Next, the closed loop gain, input resistance, and output resistance are found.

$$A = \frac{V_o}{V_s} = 9.98 \times 10^4$$

$$\beta = \frac{V_f}{V_o} = \frac{R_1}{R_1 + R_2} = 0.1$$

$$R_i = R_{id} + (R_1 || R_2) = 10 \text{ Meg}$$

$$R_o = R_A || (R_1 + R_2) = 19.7 \text{ } \Omega$$

The series mixing causes the input resistance to be increased by $D = 1 + A\beta$ and the voltage or shunt sampling of the output causes the output resistance to be reduced by a factor of D.

$$A_{vf} = \frac{A_v}{D} = 9.999$$

$$R_{if} = R_i D = 9.979 \times 10^{10} \text{ } \Omega$$

$$R_{of} = \frac{R_o}{D} = 2.0 \times 10^{-3} \text{ } \Omega$$

As a check, an exact analysis was performed on the original circuit with the following results.

$$A_{vf} = 9.999$$

$$R_{if} = 9.961 \times 10^{10} \text{ } \Omega$$

$$R_{of} = 2.0 \times 10^{-3} \text{ } \Omega$$

The agreement between feedback analysis and exact analysis is excellent. This is because the circuit meets the basic feedback assumptions very well. The feedback network is not unilateral, but this results in a very small error.

III. EXAMPLE 2

As a second example, consider the three-stage BJT amplifier shown in Figure 5. At midband, all capacitors in this circuit can be considered short circuits. Note that if the output voltage is shorted, the feedback is not removed. For this amplifier, the collector current of Q_3 is sampled by R_{10} and a current is fed back in shunt to the base node of Q_1 . Thus, this amplifier is an example of current (series) sampling and current (shunt) mixing. This feedback topology optimizes a current amplifier. The output resistance will be increased by the series sampling of collector current of the last stage, and the

input resistance will be decreased by the shunt mixing of the feedback current. Note that the load current, i_L , is not sampled directly. However, i_L is directly proportional to the sampled i_{C3} . Current sampling is often harder to analyze than voltage sampling. This is because the current sampling resistor and the load resistor are usually not directly in series. Typically, one side of the load resistor is on ground, and if the current sampling resistor is in series, it would have to be floating. It is desirable for both resistors to have one side at ground. In this example, the sampling resistor is at the collector, so it can be referenced to ac ground. Figure 6 shows the circuits necessary to solve for the open loop current gain and the feedback factor. To account for the loading of the output loop of the A circuit, the input mixing node is shorted to ground. To account for the loading of the input node of the A circuit, the output loop is severed at the collector. Thus, the A circuit has R_9 in parallel with R_F at the collector of Q_3 and the series combination of R_F and R_9 from the base to ground. The β circuit consists of a current divider involving R_F and R_9 . Since the mixing is shunt, the mixing node (base of Q_1) is shorted to ground. The open loop gain and feedback factors are defined as follows. The current gain, A_{il} , will be found first since it is the one directly controlled by the feedback loop. The actual current gain, A_i , can then be easily found from A_{il} .

Analysis of the circuits of Figure 4 yields:

$$A_{il} = \frac{i_{C3}}{i_s} = 6.1 \times 10^4$$

$$\beta = \frac{i_f}{i_{C3}} = \frac{R_{10}}{R_{10} + R_F} = 8.23 \times 10^{-3}$$

$$D = 1 + A_{il}\beta = 502$$

The input resistance of the open loop circuit as defined in the figure is found to be:

$$R_i = R_7 || R_9 || r_{\pi 1} = 2.725 \times 10^3 \Omega$$

The output resistance must be carefully chosen for the feedback analysis. The output resistance looking back from R_L is not multiplied by D since R_{11} is outside the feedback loop. Since there is current sampling, we need to break the loop in which the output current flows. We break the loop between points a and b shown in Figure 5 which is where the output current drives the load. The resistance between the parallel combination of R_{11} and R_L and the emitter of Q_3 is the open loop output

resistance. This resistance is found to be:

$$R_{ol} = R_{E3} + (R_{11} || R_L)$$

$$R_{ol} = 782.8 \Omega$$

The closed loop parameters are then found:

$$A_{ilf} = \frac{A_{il}}{D} = 121.5$$

$$R_{if} = \frac{R_i}{D} = 5.38 \Omega$$

$$R_{olf} = R_{ol}D = 393 K\Omega$$

Recall that the current gain, A_{il} , is the ratio of the collector current of Q_3 to the input current. The total current gain of the amplifier is:

$$A_{if} = \frac{i_{C3}}{i_s} \cdot \frac{i_{E3}}{i_{C3}} \cdot \frac{i_L}{i_{E3}}$$

$$A_{if} = \frac{i_L}{i_s} = A_{ilf} \cdot \frac{(1 + g_{m3}r_{\pi 3})}{g_{m3}r_{\pi 3}} \cdot \frac{R_{11}}{R_{11} + R_L}$$

$$A_{if} = 88.5$$

PSpice was used for the exact analysis of the circuit with feedback. The PSpice results for the current gain, and the input and output resistances are:

$$A_{if} = 89.1$$

$$R_{if} = 5.42 \Omega \quad R_{olf} = 396 K\Omega$$

The output resistance seen by the load resistor is:

$$R_{of} = R_{olf} || R_{11} \approx R_{11} = 2.7 K\Omega$$

The agreement between the feedback analysis and the exact analysis is quite good, but not as good as in the first example. One reason is that this circuit does not fit the feedback block diagram model as well as the first example. For example, i_{C3} is sampled rather than i_L . Another source of error is that for simplicity the analysis of the open loop amplifier in Example 2 ignored the resistor, r_o , across the controlled source in the transistor models. Also, PSpice calculates the transistor model parameters from the bias and these parameters differ from the ones estimated from the Q point and used in the

open loop analysis.

It should be noted that the usefulness of feedback analysis is not to obtain great accuracy. The main advantage of feedback analysis is to get a good approximate answer easily and, furthermore, obtain a simple expression for important parameters such as gain input and output resistances. In the second example, the gain of the amplifier can be approximated by:

$$A_{if} = A_{il} \cdot \frac{i_{E3}}{i_{C3}} \cdot \frac{i_L}{i_{E3}}$$

$$A_{if} = \frac{1}{\beta} \cdot \frac{(1 + g_{m3} r_{\pi 3})}{g_{m3} r_{\pi 3}} \cdot \frac{R_{11}}{R_{11} + R_L}$$

$$\beta = \frac{R_{10}}{R_{10} + R_F}$$

Similarly, the expressions for the input and output resistances are:

$$R_{if} = \frac{R_i}{D} = \frac{R_{12} r_{\pi 1}}{R_{12} + r_{\pi 1}} \cdot \frac{1}{D}$$

$$R_{oif} = R_{oi} D = \left(\frac{r_{\pi 3} + (R_7 || R_9)}{1 + g_{m3} r_{\pi 3}} + (R_{11} || R_L) \right) \cdot D$$

It would be impractical to try to derive analytical expressions using circuit analysis for the exact closed loop of such an amplifier. The above expressions, however, can be found simply and are easy to interpret. These expressions are quite useful when designing the circuit. Once the circuit is designed, a circuit simulator such as PSpice can be used to fine tune the design.

IV. SUMMARY

A methodology for applying feedback analysis to a feedback amplifier has been presented. The two examples were chosen to illustrate both types of mixing and both types of sampling. The first example met the basic assumptions very well and conformed to the feedback block diagram and, therefore, feedback analysis was nearly exact. The circuit in the second example did not satisfy these criteria as well, but did agree well enough for engineering design purposes.

It was shown that feedback analysis is an approximate technique that can be of significant use for the analysis and design of multistage amplifiers. When

using feedback analysis, it must be kept in mind that the circuit must conform to the feedback block diagram as much as possible, and the basic feedback assumptions must be met. Not only is feedback analysis useful for multistage amplifiers, but it is actually easier to apply and is more accurate than for most single stage amplifiers. This is because large gain multistage amplifiers better fit the basic assumptions that the A and β networks do load each other and are nearly unilateral. Furthermore, most single stage amplifiers, even if the criteria for applying feedback analysis are satisfied, are easier to solve directly than by using feedback analysis. Students should be taught to first satisfy the basic assumptions to determine when feedback analysis is appropriate, and how to relate the circuit to be analyzed to the feedback block diagram. They should also learn to use the simple approximate expressions as a tool for initial design and for circuit refinement using a circuit analyzer such as PSpice.

REFERENCES

- [1] *MICROELECTRONICS - 2ND ED.*, Millman and Grabel, McGraw-Hill, New York, 1987.
- [2] *MICROELECTRONIC CIRCUITS 4TH ED.*, Sedra and Smith, Oxford University Press, New York, 1998.

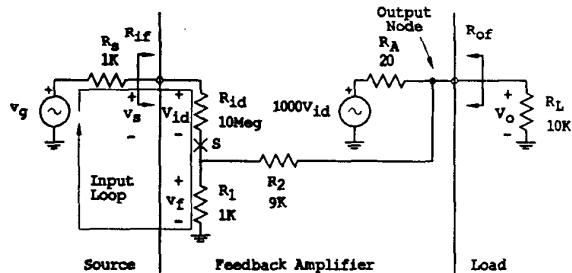


Figure 3. Non-inverting Op Amp Circuit for Example 1.

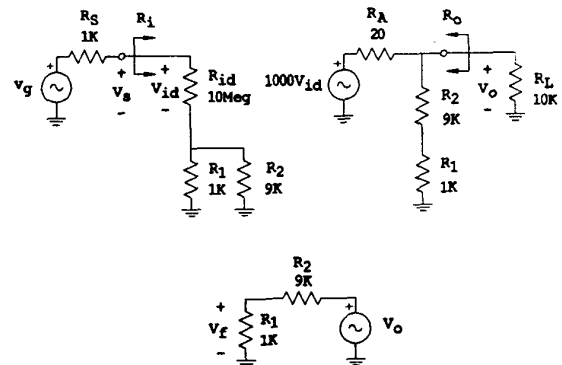
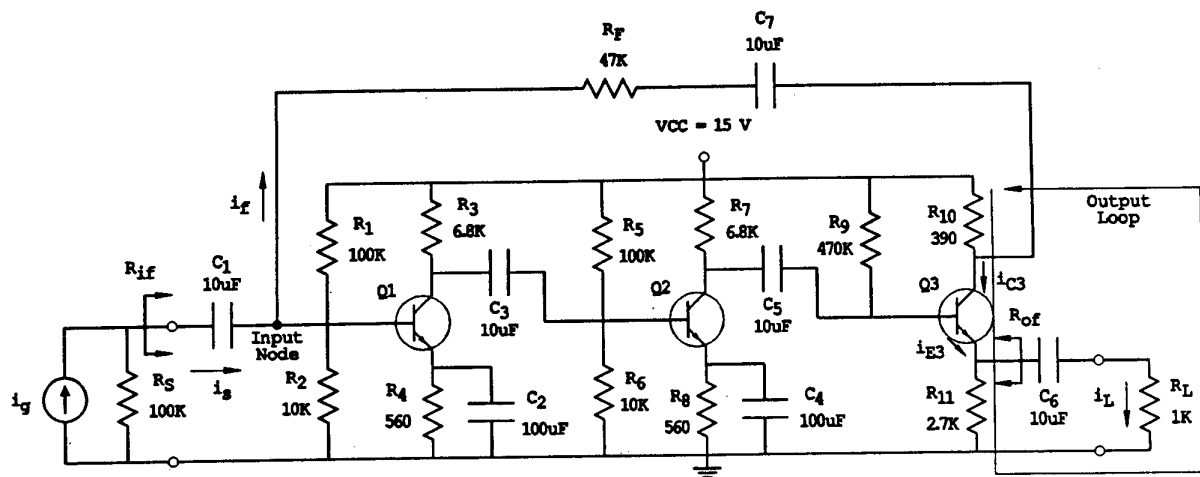


Figure 4. A and β Circuits for Example 1.



Transistor Q1

$I_{C1} = 1 \text{ mA}$
 $V_{CE1} = 7.5 \text{ V}$
 $g_{m1} = 40 \text{ mA/V}$
 $r_{\pi1} = 3.75 \text{ K}$

Transistor Q2

$I_{C2} = 1 \text{ mA}$
 $V_{CE2} = 7.5 \text{ V}$
 $g_{m2} = 40 \text{ mA/V}$
 $r_{\pi2} = 3.75 \text{ K}$

Transistor Q3

$I_{C3} = 2.4 \text{ mA}$
 $V_{CE3} = 7.5 \text{ V}$
 $g_{m3} = 96 \text{ mA/V}$
 $r_{\pi3} = 1.56 \text{ K}$

Figure 5. Three Stage BJT Feedback Amplifier for Example 2.

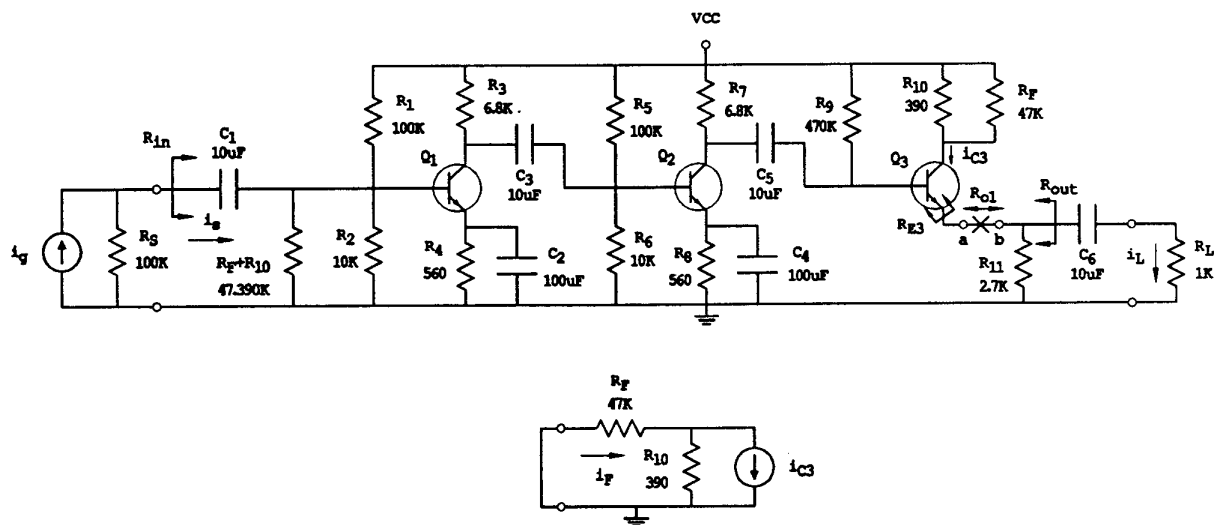


Figure 6. A and β Circuits for Example 2.